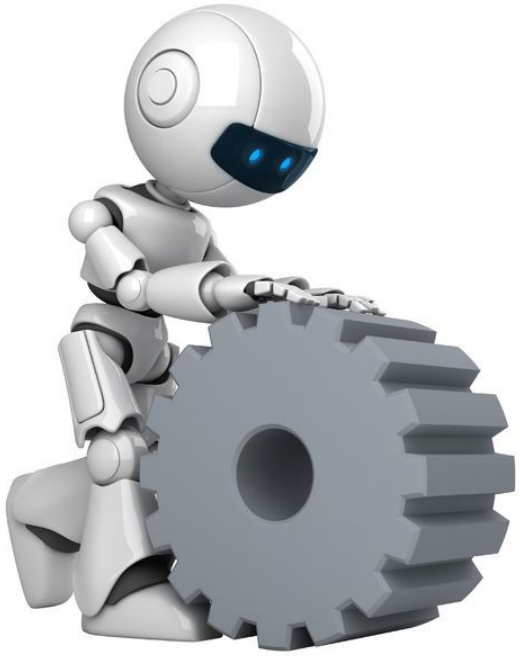
ICAT3110-3011

Intelligent Robotics



Lecture 6: Perception II

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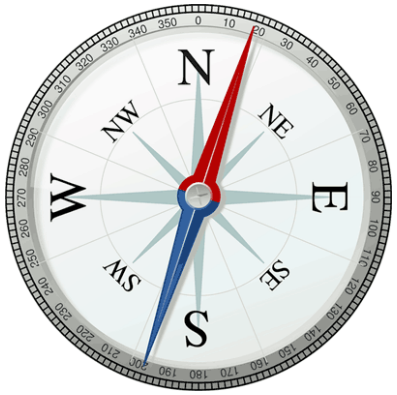
Sensor Performance



- **Cross-sensitivity** is the technical term for **sensitivity** to **environmental** parameters that are orthogonal to the **target parameters** for the sensor.

- **For example**, a flux-gate **compass** can demonstrate **high sensitivity** to magnetic **north** and is therefore of use for mobile robot navigation.

- However, the compass will also demonstrate **high sensitivity** to **ferrous building** materials, so much so that its **cross-sensitivity** often makes the sensor **useless** in some **indoor environments**.

- **High** cross-sensitivity of a sensor is generally **undesirable**, especially when it cannot be **modeled.**

Sensor Performance

- **<u>Error</u>** of a sensor is defined as the **difference** between the sensor's **output measurements** and the **true values** being **measured**, within some specific **operating** context.

- Given a true value **v** and a measured value **m**, we can define error as

$$\boxed{error = m - v}$$

- **<u>Accuracy</u>** is defined as the **degree** of **conformity** between the sensor's **measurement** and the **true** value, and is often **expressed** as a **proportion** of the **true** value (e.g., 97.5% accuracy).

- Thus **small error** corresponds to **high accuracy** and vice versa.

$$\boxed{Accuracy = 1 - \frac{|error|}{v}}$$

Sensor Performance

- Of course, obtaining the **ground** truth, **v**, can be difficult or impossible, and so establishing a confident characterization of **sensor accuracy** can be **problematic**.
- Further, it is important to distinguish between **two different sources of error**:
- Systematic errors are caused by **factors** or **processes** that can in theory be **modeled**.
- These errors are, therefore, **deterministic** (i.e., predictable).
- **e.g.** Poor calibration of a laser rangefinder, an unmodeled slope of a hallway floor, and a bent stereo camera head due to an earlier collision are all **possible causes** of systematic sensor errors.
- Random errors **cannot** be **predicted** using a sophisticated **model** nor can they be **mitigated** by more precise sensor machinery.
- These errors can only be described in **probabilistic terms** (i.e., **stochastically**).
- **e.g.** Hue instability in a color camera, spurious range-finding errors, and black level noise in a camera are all **examples** of **random** errors.

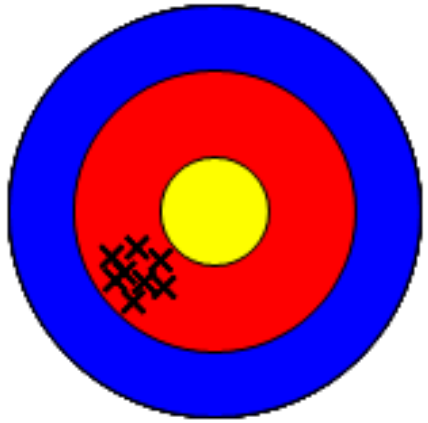
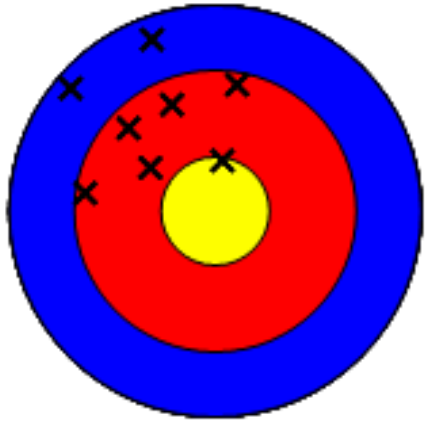
Sensor Performance

- **Precision** is often confused with **accuracy**, and now we have the tools to clearly distinguish these two **terms**. Intuitively, **high precision** relates to reproducibility of the **sensor results**.
- **For example**, one sensor taking **multiple** readings of the **same** environmental state has **high precision** if it produces the **same output**.
- Another example, **multiple copies** of this sensor taking **readings** of **same** environmental state have **high precision** if their outputs **agree**. Precision does not have any bearing on the **accuracy** of the sensor's **output** with **respect** to the true value being measured.
- Suppose that the **random error** of a sensor is characterized by some mean value $\boldsymbol{\mu}$ and a standard deviation $\boldsymbol{\sigma}$.
- The formal **definition** of **precision** is the **ratio** of the sensor's **output range** to the **standard deviation**

$$\boxed{Precision = \frac{range}{\sigma}}$$

- Only σ has impact on precision.
- But μ is proportional to overall sensor error and inversely proportional to sensor accuracy.

Sensor Performance

	Accurate	Inaccurate (systematic error)
Precise		
Imprecise (reproducibility error)		

Sensor Performance

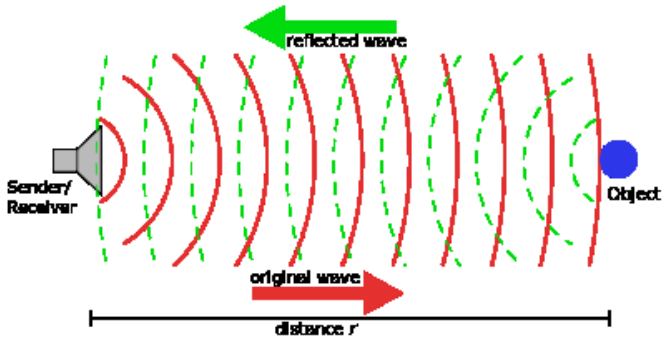
	Results A	Results B	Results C
Accuracy	Low	High	Low
Precision	High	High	Low
Systematic error	High	None	No
Random Error	No	None	High

Challenges in Mobile Robots

- Mobile robots depend heavily on **exteroceptive** sensors, which concentrate on acquiring information on **from** the surroundings. **In mobility** case, their **ever-changing position** and their motion have a significant **impact** on overall **sensor behavior**.

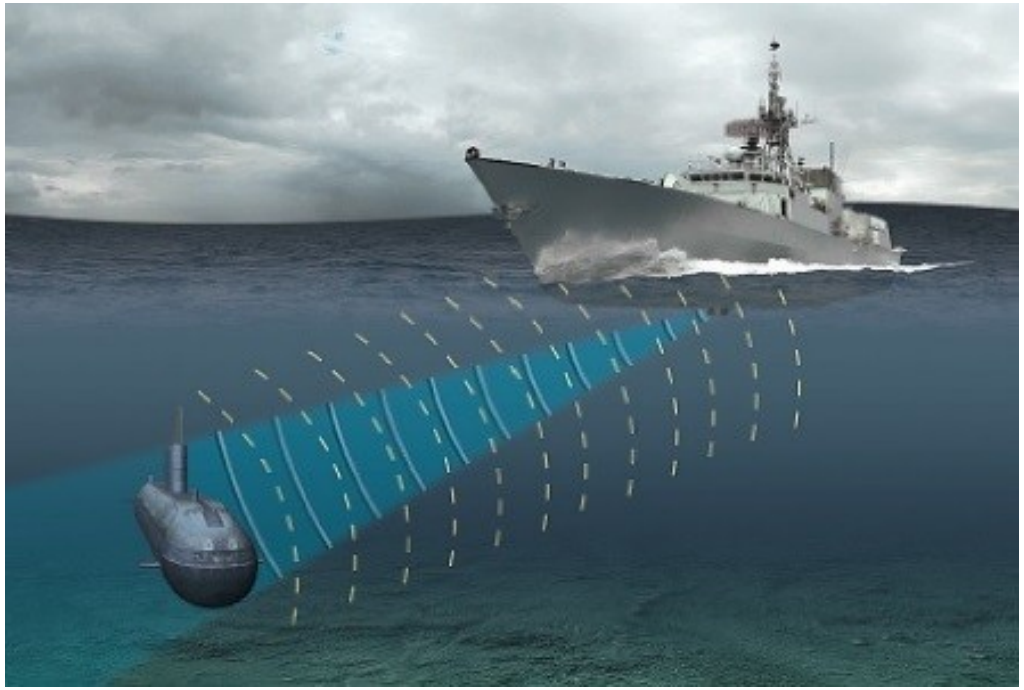
Blurring of systematic and random errors

- Active **ranging sensors** tend to have **failure modes** that are triggered largely by specific relative **positions** of the sensor and **environment** targets.
- For example, a **sonar** sensor will produce specular reflections, producing grossly **inaccurate** measurements of **range**, at specific **angles** to a **smooth sheetrock** wall.
- This is especially true in a mobile robot **outfitted** with a **ring** of multiple **sonars.**
- If the robot's **static** position causes a particular sonar to **fail** in this manner, the **sonar** will fail **consistently** and will tend to return **precisely** the same (and incorrect!) reading time after time.



Challenges in Mobile Robots

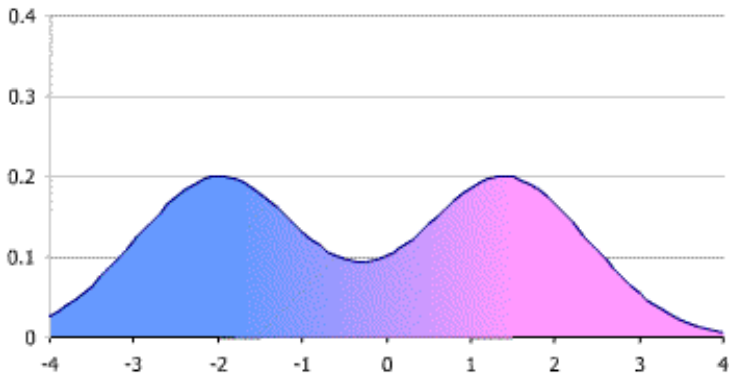
- The fundamental mechanism here is the cross-sensitivity of mobile robot **sensors** to robot **pose** and **robot-environment dynamics**.
- The models for such cross-sensitivity are not, truly random. However, these physical interrelationships are **rarely modeled** and therefore, from the point of view of an incomplete model, the **errors appear random** during motion and **systematic** when the **robot** is at rest.
- Sonar is **not** the **only** sensor subject to this **blurring** of **systematic** and **random** error modality.
- while **systematic** error and **random** error are well-defined in a controlled setting, the mobile robot can exhibit error characteristics that **bridge** the **gap** between **deterministic** and **stochastic** error mechanisms.

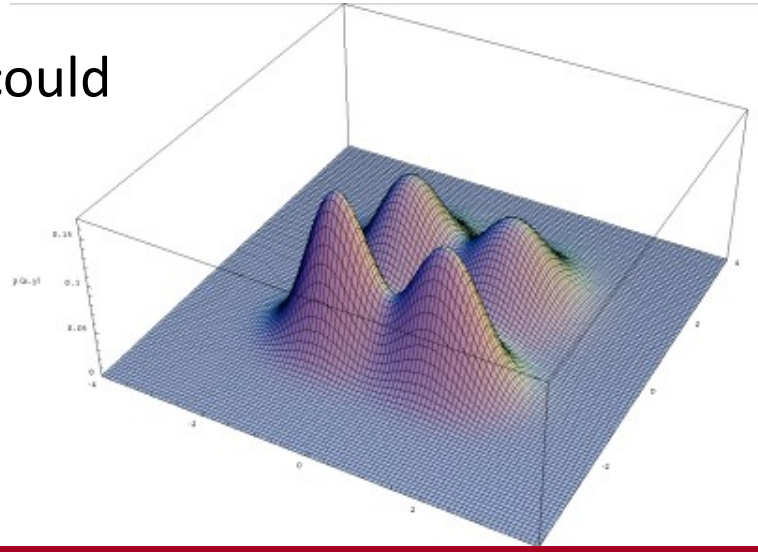


Challenges in Mobile Robots

Multimodal error distributions.

- It is common to characterize the **behavior** of a sensor's **random** error in terms of a **probability** distribution over **various output** values.
- Consider, the **sonar sensor** once again. when the sonar sensor is **moving** through an environment and is sometimes **faced** with **materials** that cause **coherent reflection** rather than **returning** the sound signal to the sonar sensor, then the sonar will **overestimate** the **distance** to the object. The probability distribution could easily be at least **bimodal.**





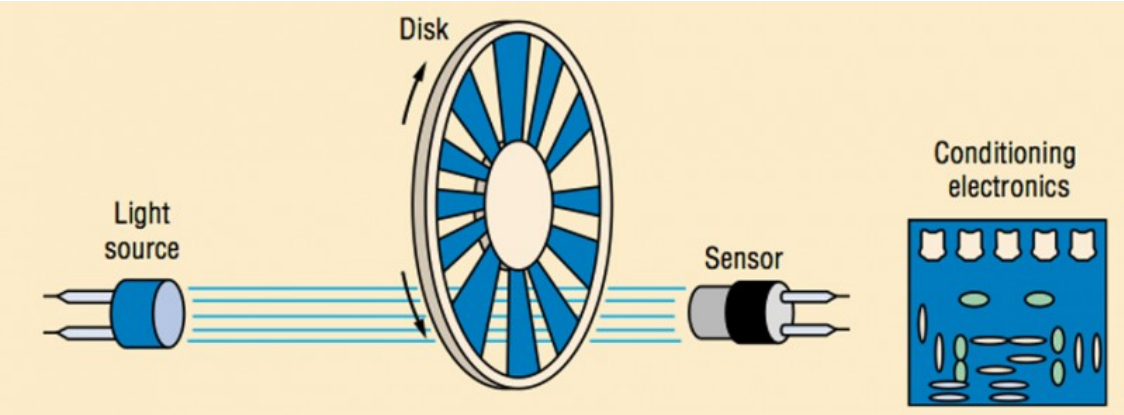
- **More generally**, a multimodal distribution is a continuous **probability distribution** with two or more modes.

Wheel/Motor Sensors

EC or PC ? Proprioceptive

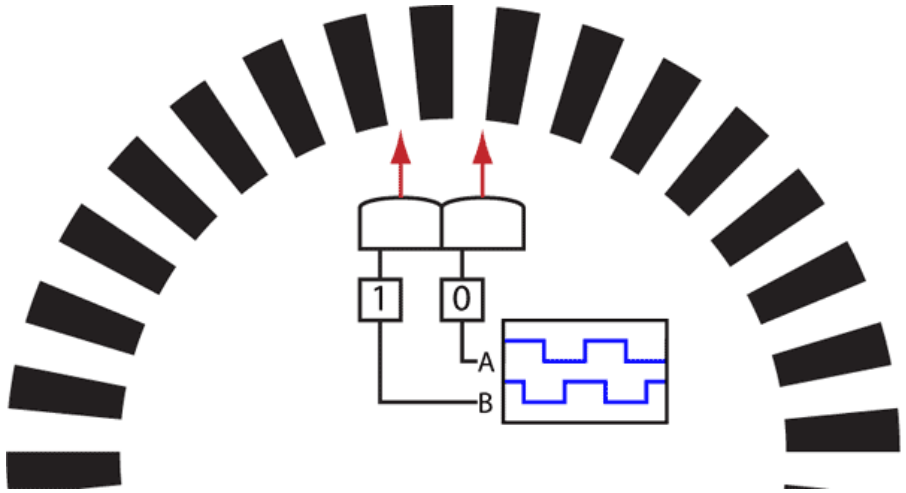
- These are **devices** used to **measure** the **internal state** and **dynamics** of a **mobile** robot.
- These sensors have **vast applications outside** of **mobile** robotics and, as a result, mobile robotics has enjoyed the benefits of **high-quality**, **low-cost** wheel and motor sensors that offer **excellent** resolution.

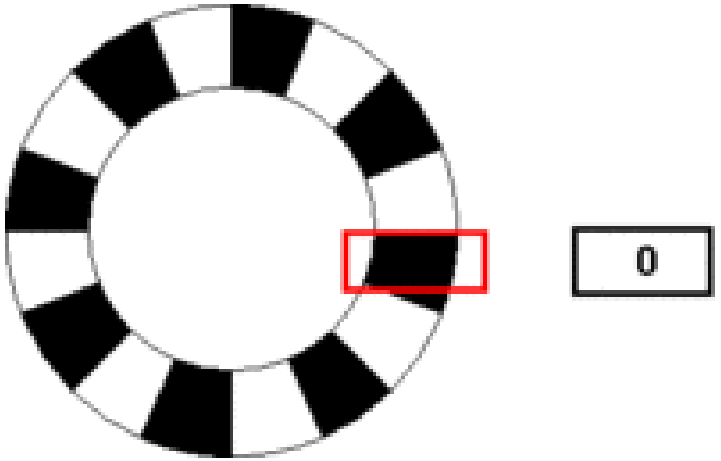
Optical Encoders

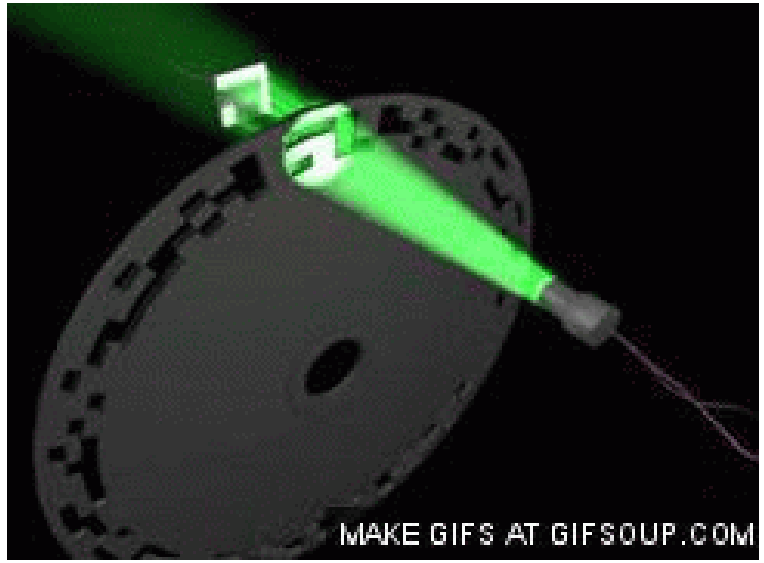


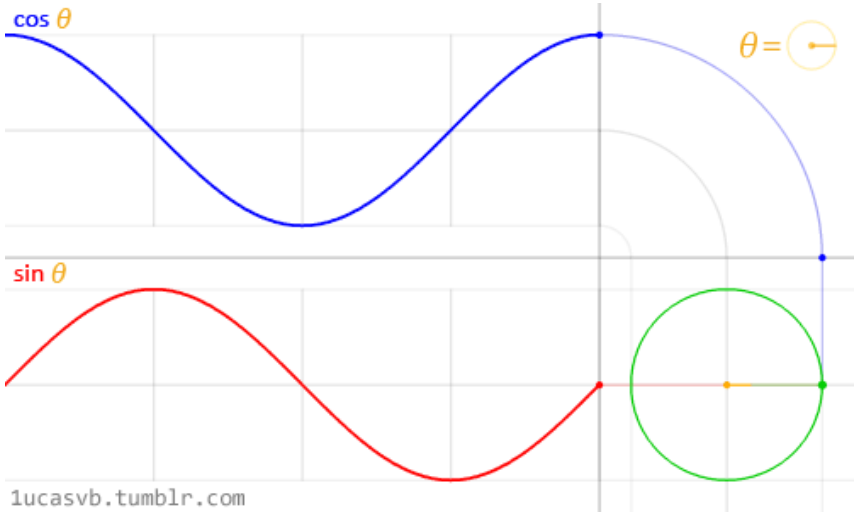
- **Mechanical light chopper** that produces a **certain** number of **sine** or **square wave** pulses for each shaft revolution.
- It **consists** of an **illumination** source, a fixed **grating** that masks the light, a **rotor** disc with a fine **optical grid** that rotates with the shaft, and **fixed optical** detectors.
- In **robotics**, the resulting **sine** wave is transformed into a **discrete square** wave using a threshold to choose between light and dark states. **Resolution** is measured in **cycles** per **revolution** (**CPR**).

Optical Encoders







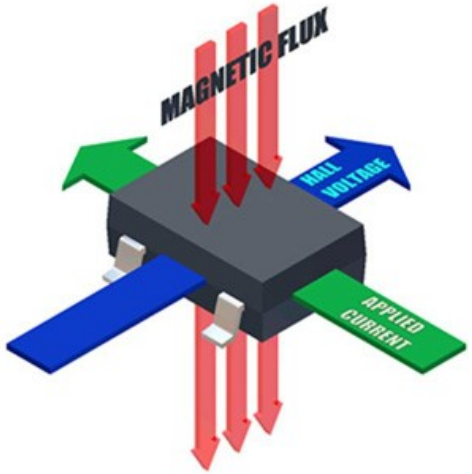


Heading Sensors

- Heading sensors can be **proprioceptive** (gyroscope, inclinometer) or **exteroceptive** (compass). They are used to determine the robot's **orientation** and inclination.
- They allow us, together with appropriate **velocity information**, to integrate the **movement** to a **position** estimate.

Compasses

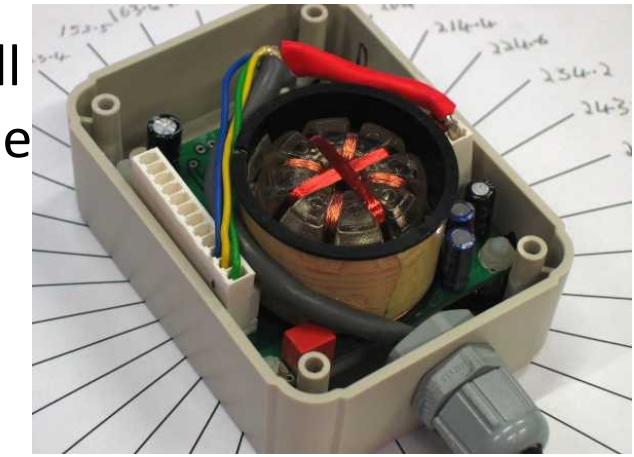
- The **two most** common **modern** sensors for measuring the direction of a **magnetic** field are the **Hall effect** and **flux gate compasses**. Each has advantages and disadvantages.



- **<u>The Hall effect</u>** describes the **behavior** of electric potential in a semiconductor when a constant current is applied across the length of a semiconductor, there will be a **voltage difference** in the **perpendicular** direction, across the semiconductor's width, based on the relative orientation of the semiconductor to magnetic flux lines.

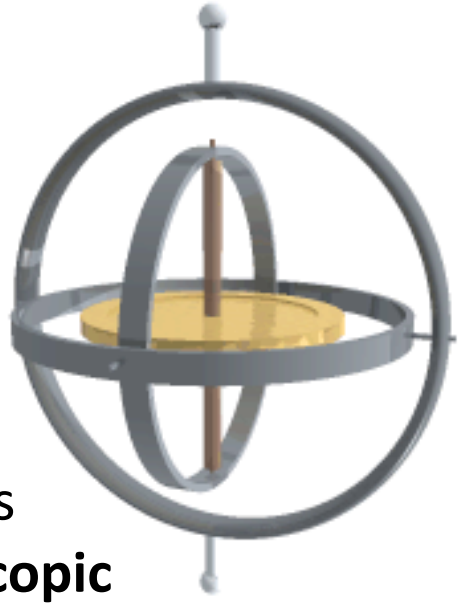
Compasses

- **Hall effect** digital compasses are **popular** in mobile robotics, and **inexpensive**. Also suffer from a range of **disadvantages**. **Resolution** of a digital Hall effect compass is **poor**. **Internal** sources of **error** include the **nonlinearity** of the **basic** sensor and **systematic** bias errors at the semiconductor level.
- **The flux gate** compass operates on a different principle. Two small coils are wound on ferrite cores and are fixed perpendicular to one another.



- The **difference** between a **magnetic** compass and an **electronic** compass is that in the former variety there is a pointer that constantly moves indicating the direction.
- However, in an electronic compass there are **no pointers** that specify the **direction**.
- **Electric currents** that **pass** through **coils** of wire that are kept **inside** the compass **indicate** the **geographic direction** through signals that are displayed **digitally**.
- **Disadvantage**: If **Electricity-OFF** on the **ship** then the device **will not function .**

Gyroscopes



- Gyroscopes are **heading** sensors which **preserve** their **orientation** in relation to a fixed **reference** frame. Thus they provide an **absolute measure** for the **heading** of a **mobile** system.

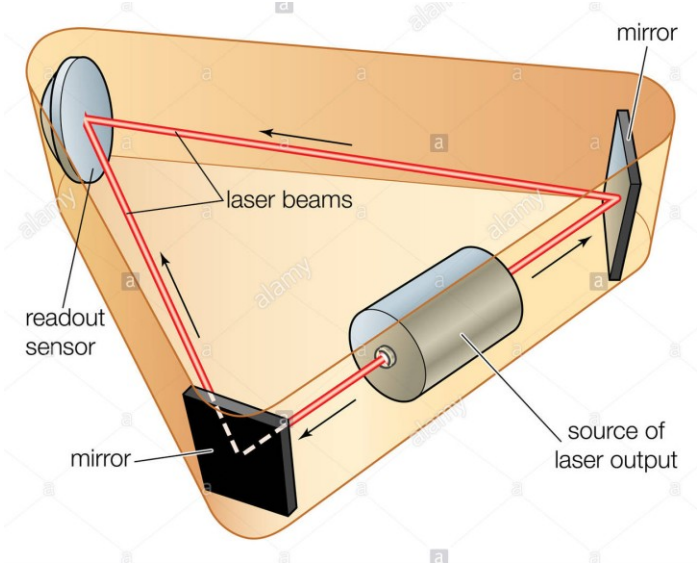
Mechanical Gyroscopes

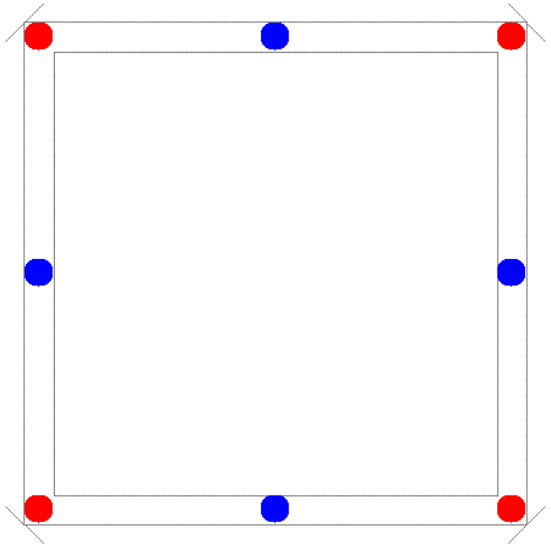
- The concept of a **mechanical** gyroscope relies on the **inertial** properties of a **fast-spinning** rotor. The property of interest is known as the **gyroscopic precession**. If you try to rotate a **fast-spinning** wheel around its **vertical** axis, you will feel a harsh **reaction** in the **horizontal** axis.
- This is due to the **angular momentum** associated with a **spinning wheel** and will keep the axis of the gyroscope inertially **stable**.
- The reactive **torque** τ and thus the **tracking** stability with the **inertial** frame are proportional to the **spinning** speed $\boldsymbol{\omega}$, the precession speed $\boldsymbol{\Omega}$, and the wheel's inertia **I**.

$$\boxed{\tau = I\omega\Omega}$$

Optical Gyroscopes

- **Optical** gyroscopes are a relatively **new innovation**.
- **Commercial** use began in the early 1980s when they were first installed in **aircraft**. Optical gyroscopes are **angular speed** sensors that use two **monochromatic** light beams, or **LASERs**, emitted from the same source, **instead** of moving, **mechanical** parts.
- They work on the **principle** that the **speed** of light **remains unchanged** and, therefore, geometric change can cause light to take a varying **amount** of time to reach its **destination**.
- **One laser** beam is **sent** traveling **clockwise** through a fiber while the other **travels counterclockwise**.
- Because the laser **traveling** in the **direction** of **rotation** has a slightly **shorter** path, it will have a **higher frequency**.

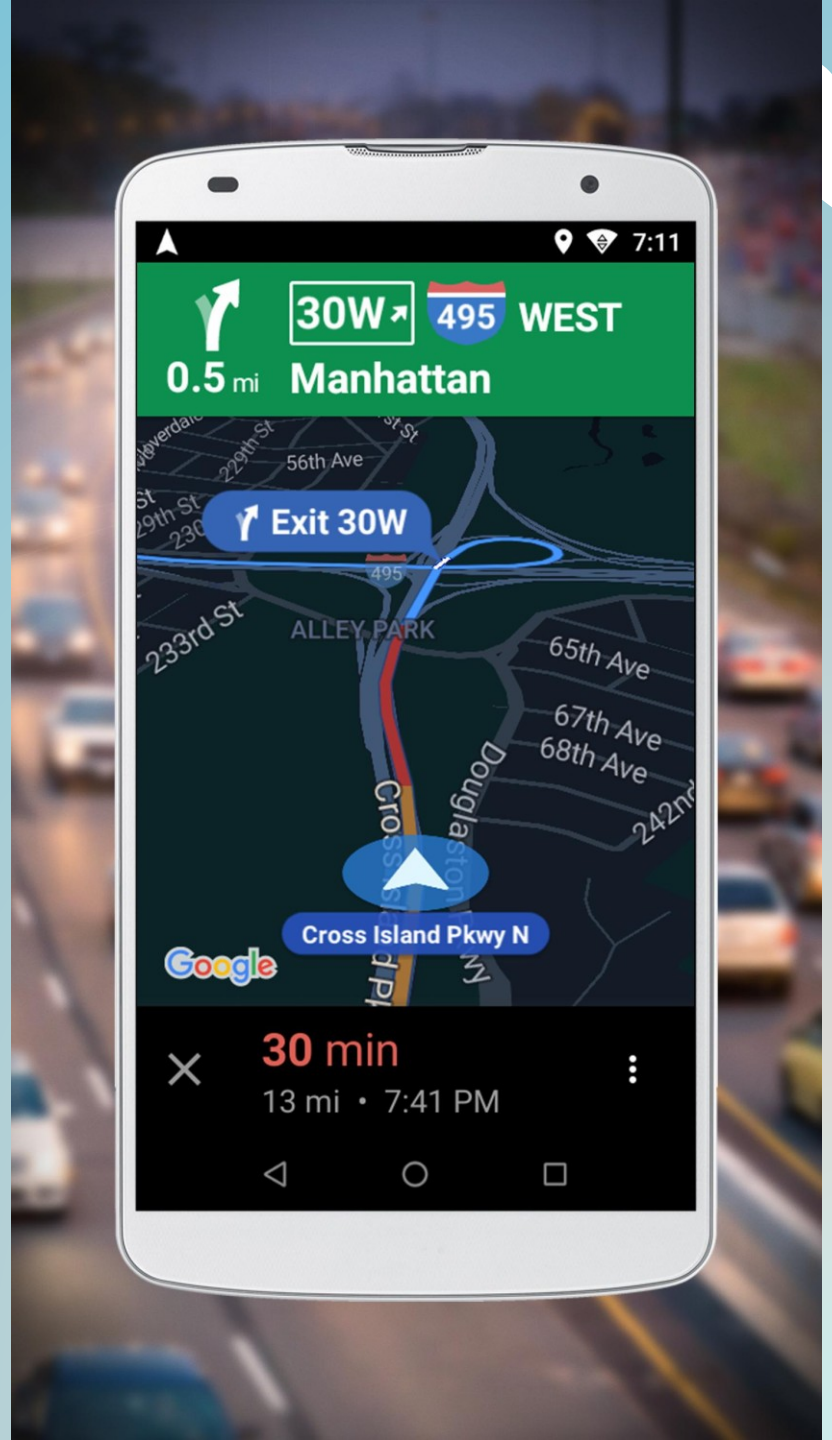




Mobile Robot Localization

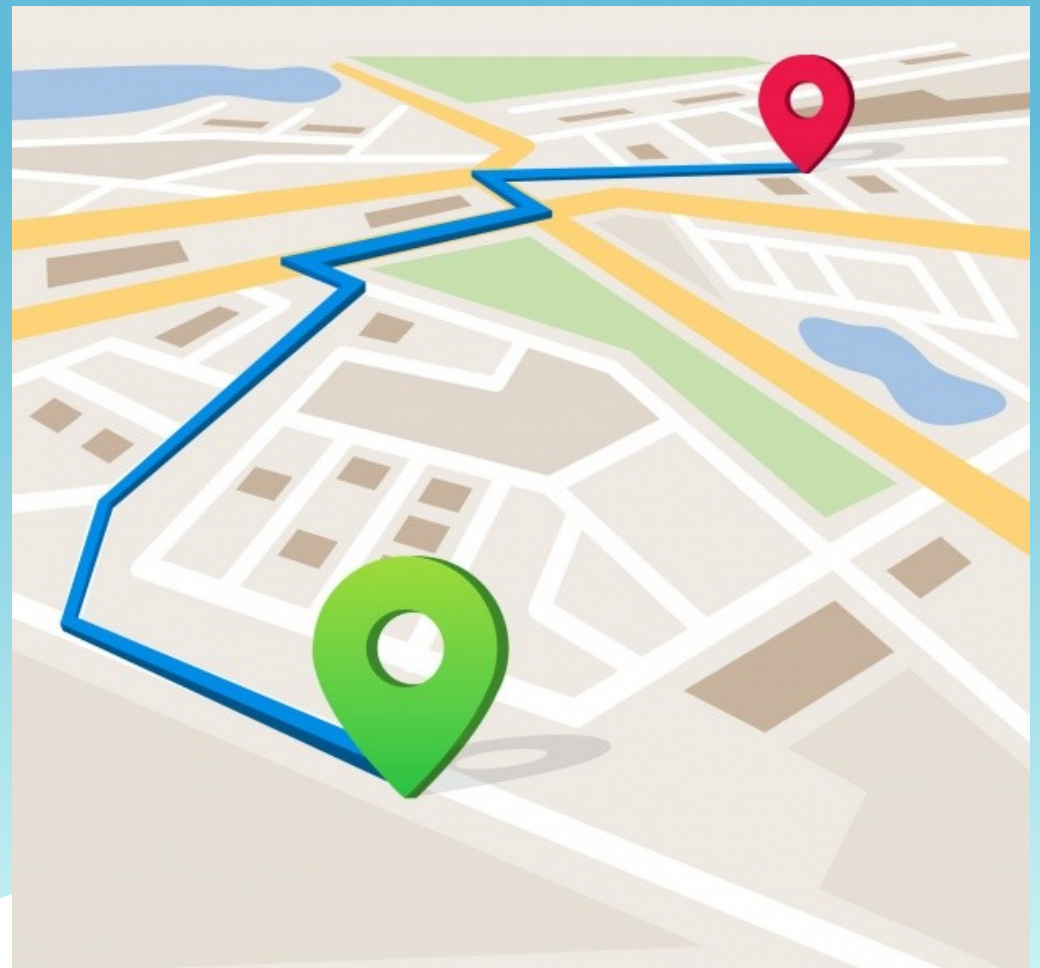
Introduction – Navigation

- Turn right after 300 meters
- continue for 5 kilometers
- turn left after 100 meters
- you ‘ve arrived to destination



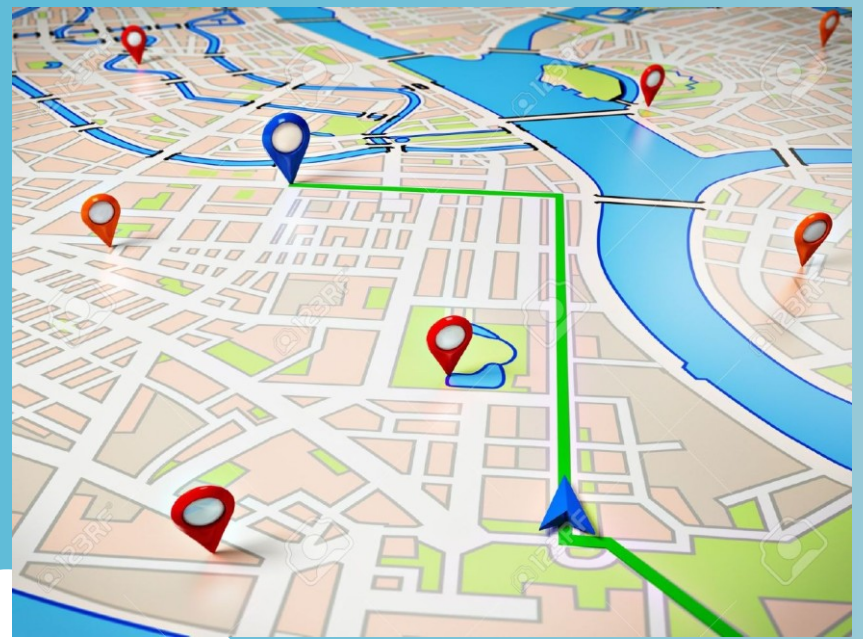
Navigator in ancient times

The World Before GPS



Destination

1. We want to reach our Destination
2. Destination requries Navigation
3. Navigation requires Localization



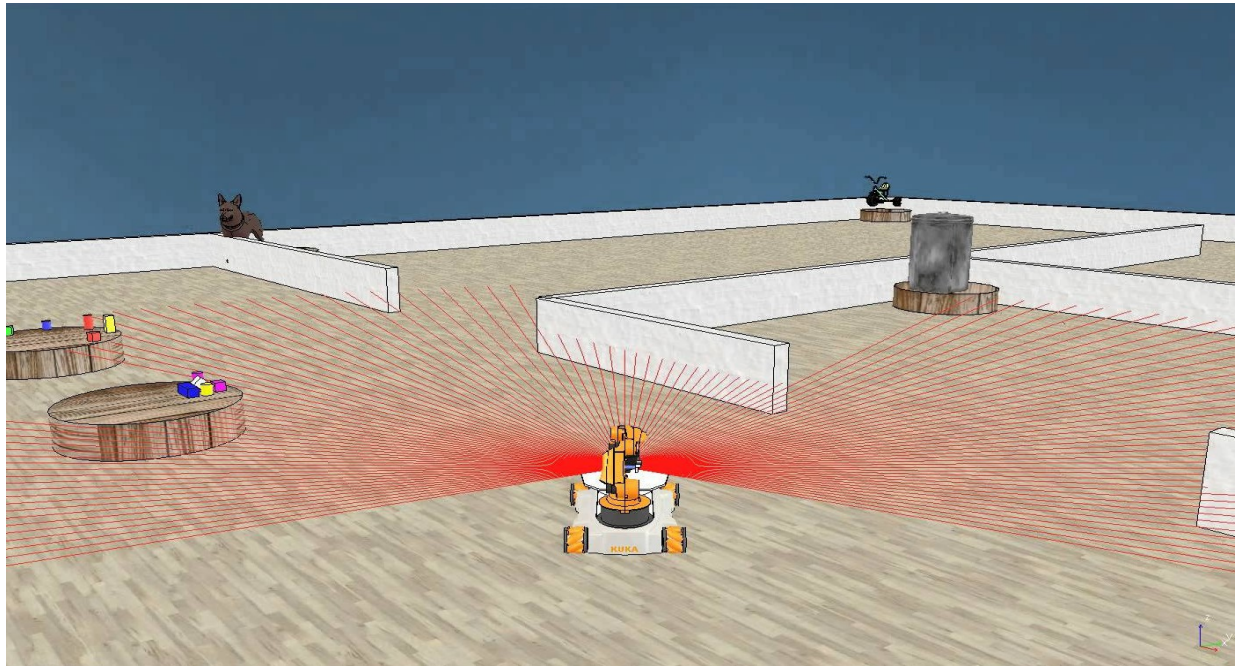
Localization (or Positioning)

Introduction – Navigation

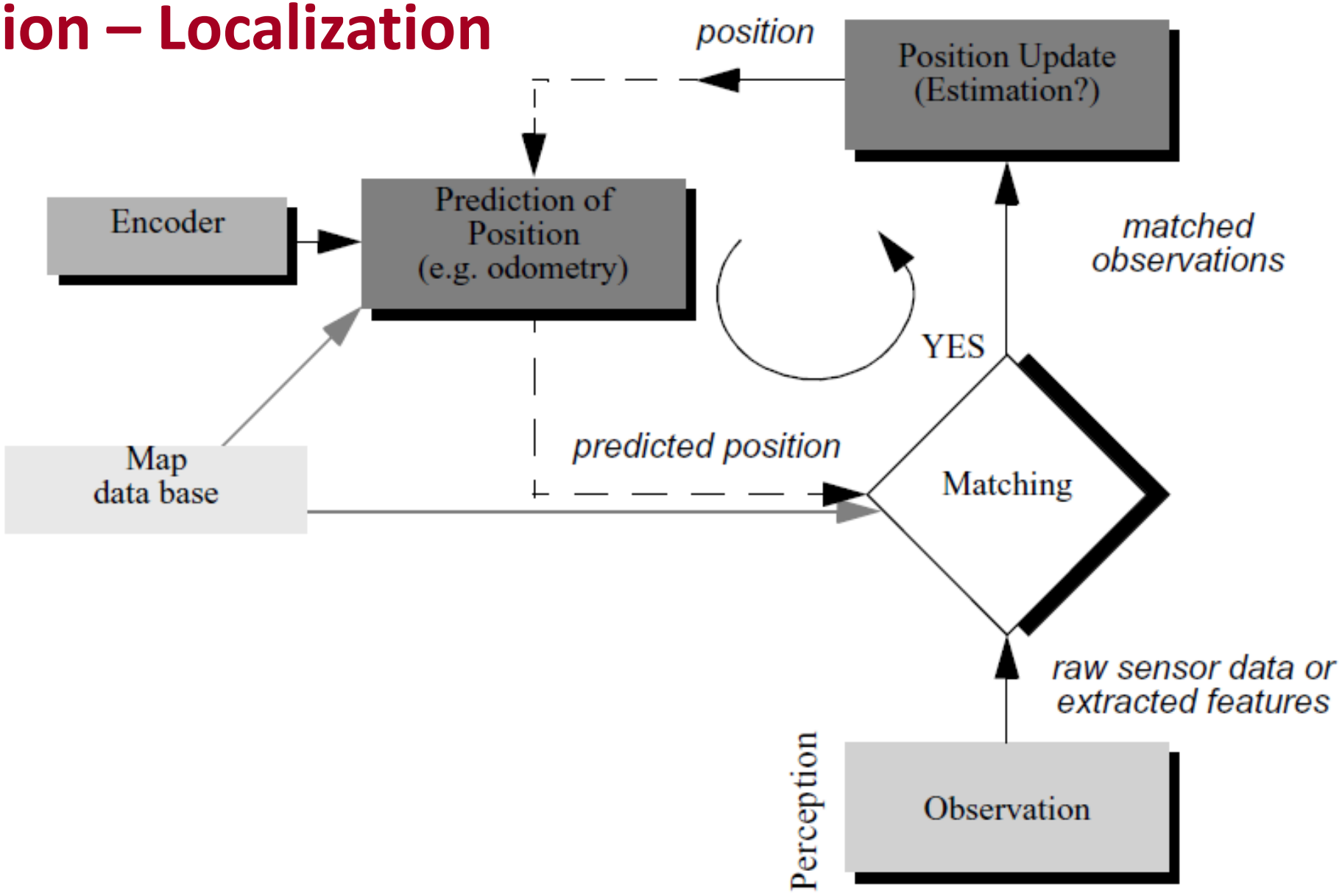
- **Navigation** is one of the most **challenging** competences required of a mobile robot.
- Success in navigation requires success at the **four** building blocks of navigation:

1. **Perception**, the robot must **interpret** its **sensors** to extract **meaningful** data;
2. **Localization**, the robot must **determine** its **position** in the environment ;
3. **Cognition**, the robot must **decide how** to **act** to **achieve** its goals;
4. **Motion control**, the robot must **modulate** its **motor outputs** to achieve the **desired** trajectory.

- Of these four components, **Localization** has received the **greatest research** attention in the **past** decade and, as a result, **significant** advances have been made on this front.



Introduction – Localization

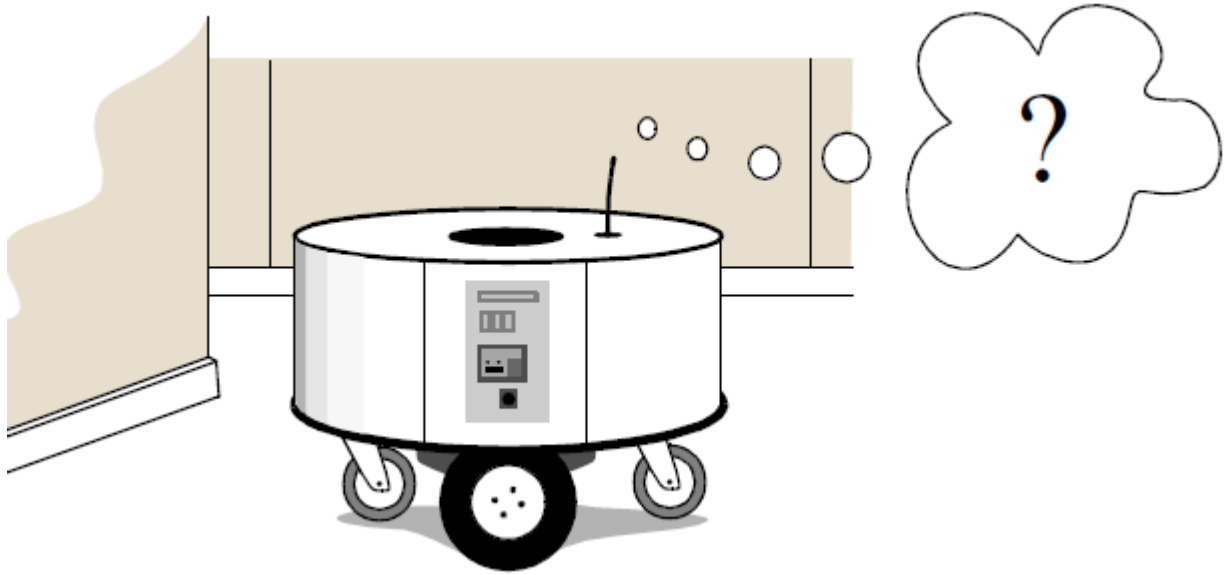


Bonus question

What is the Difference between Navigation and Localization

Introduction – Localization

- If one could attach an accurate **GPS** (global positioning system) sensor to a mobile robot, **much** of the **localization problem** would be **solved**.
- The GPS would **inform** the robot of its **exact position**, indoors and outdoors, so that the answer to the question, "**Where am I**?", would always be **immediately available**.
- Though, such a sensor is **not currently practical**. The GPS network provides **accuracy** to within **several meters**, which is **unacceptable** for **localizing** some mobile robots.
- Furthermore, GPS technologies **cannot** function **indoors** or in **obstructed** areas and are thus **limited** in their **workspace**.
- But, **looking beyond** the **limitations** of GPS, **localization** implies more than **knowing** one's **absolute position** in the Earth's **reference** frame.

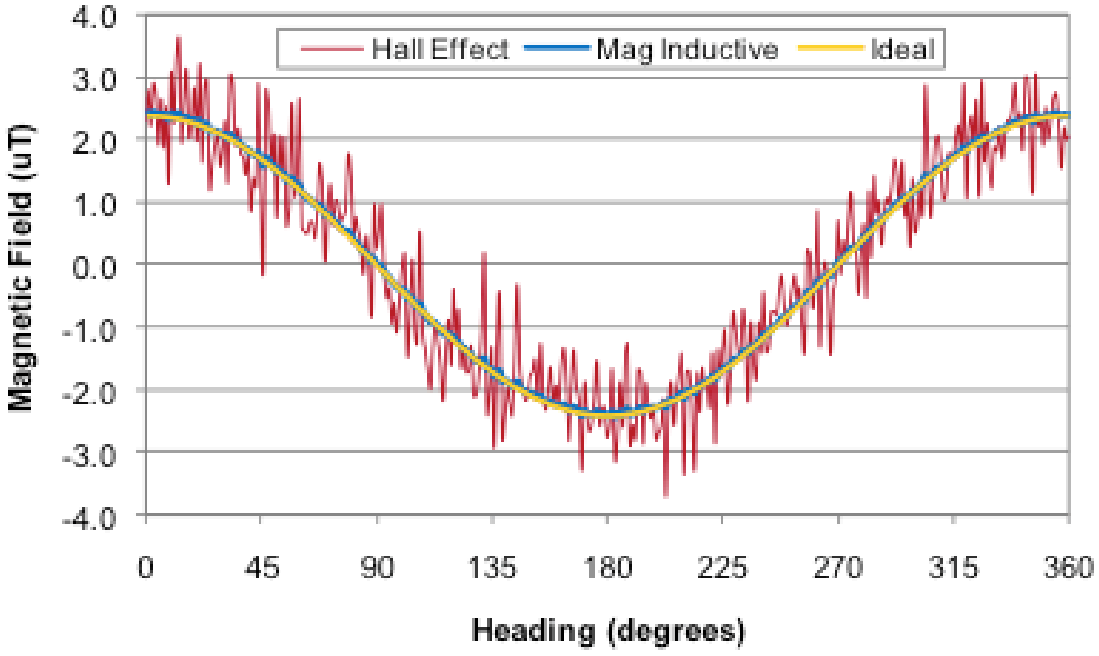


Localization challenges

- Consider a robot that is **interacting** with **humans**.
- This robot may need to **identify** its **absolute position**, but its **relative position with** respect to **target humans** is equally **important**.
- Its **localization** task can include **identifying humans** using its **sensor** array, then computing its **relative position** to the humans.
- Furthermore, during the **cognition** step a robot will **select a strategy** for **achieving** its goals.
- If it intends to reach a **particular location**, then **localization may not** be **enough**. The robot may need to **acquire** or build an **environmental model**, a **map**, that aids it in planning a path to the goal.

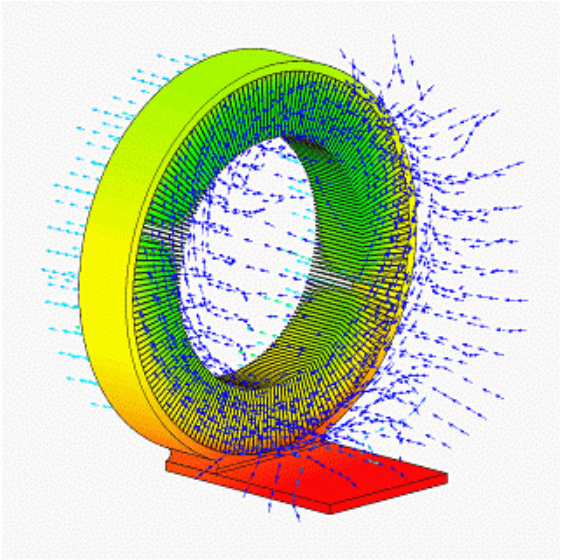
Localization challenges - Sensor noise

- Sensor noise **induces** a **limitation** on the **consistency** of sensor **readings** in the same **environmental** state and, therefore, on the number of **useful bits available** from each **sensor reading**.
- Often, the source of sensor noise **problems** is that some **environmental features** are **not captured** by the robot's **representation** and are thus **overlooked**.
- Sensor noise is **an undesirable** content that **reduces** the **useful information** content of sensor readings.
- Clearly, the **solution** is to take **multiple readings** into account, employing temporal **fusion** or **multi-sensor** fusion to **increase** the **overall information** content of the robot's inputs.

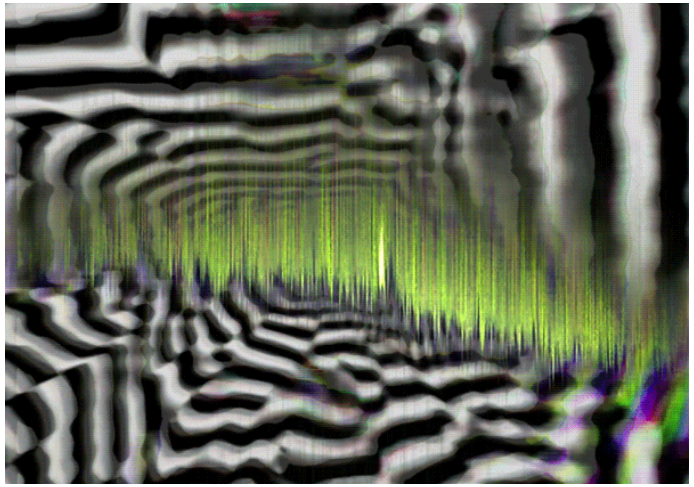


Localization challenges - Sensor noise

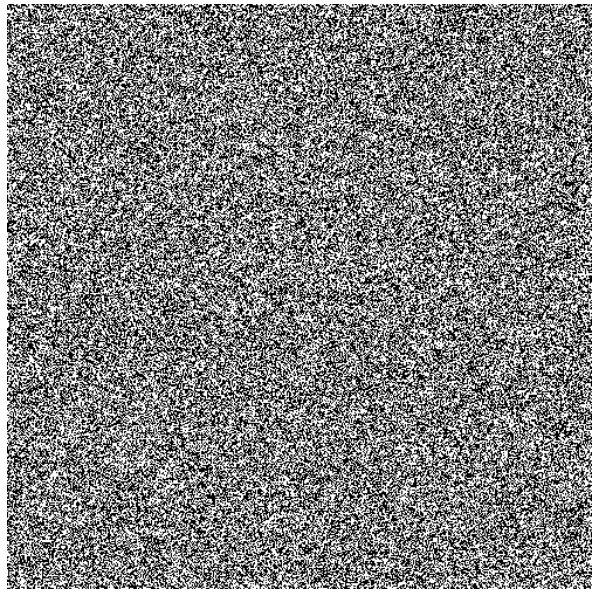
Thermal Noise



Flickr Noise



Shot Noise

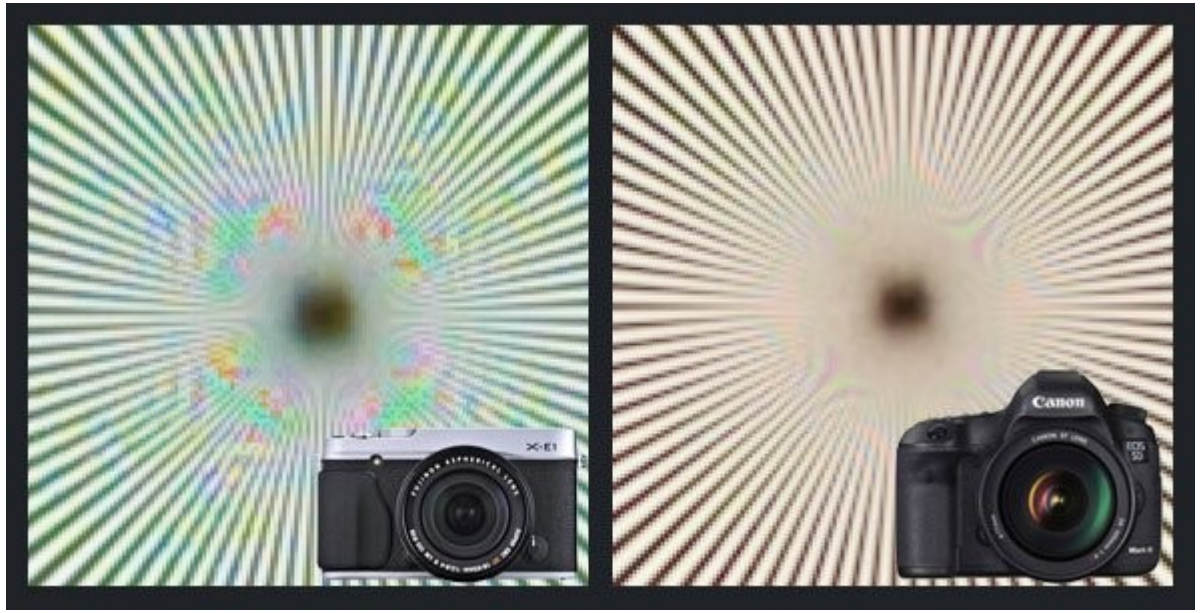


Localization challenges - Sensor Aliasing

- Sensor **aliasing** is a phenomenon that **humans rarely encounter**.
- The human **sensory system**, particularly the **visual** system, tends to receive **unique inputs** in each **unique local** state.
- In other words, every different place **looks different.** Consider **moving through** an **unfamiliar** building that is completely **dark**. When the **visual** system sees only **black**, one's **localization** system quickly **degrades**.

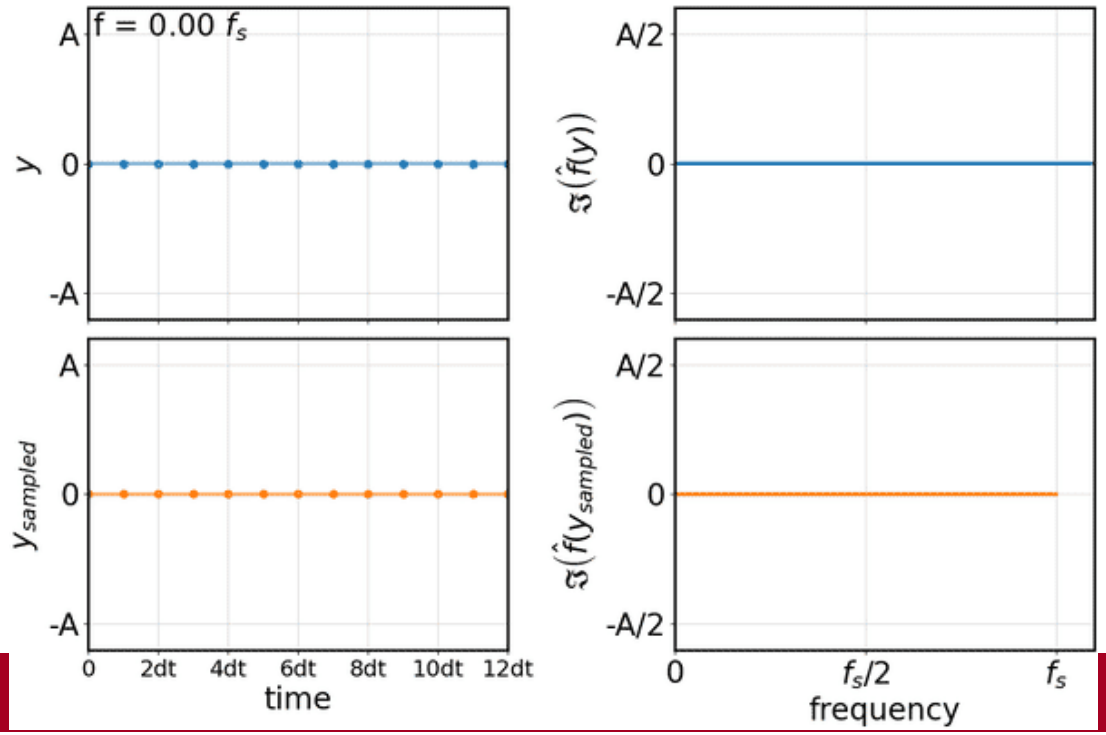
- In robots, the **non-uniqueness** of sensor readings, or sensor aliasing, is the **norm** and **not the exception**.

- A classic problem with sonar-based robots involves distinguishing between humans and inanimate objects in an indoor setting.

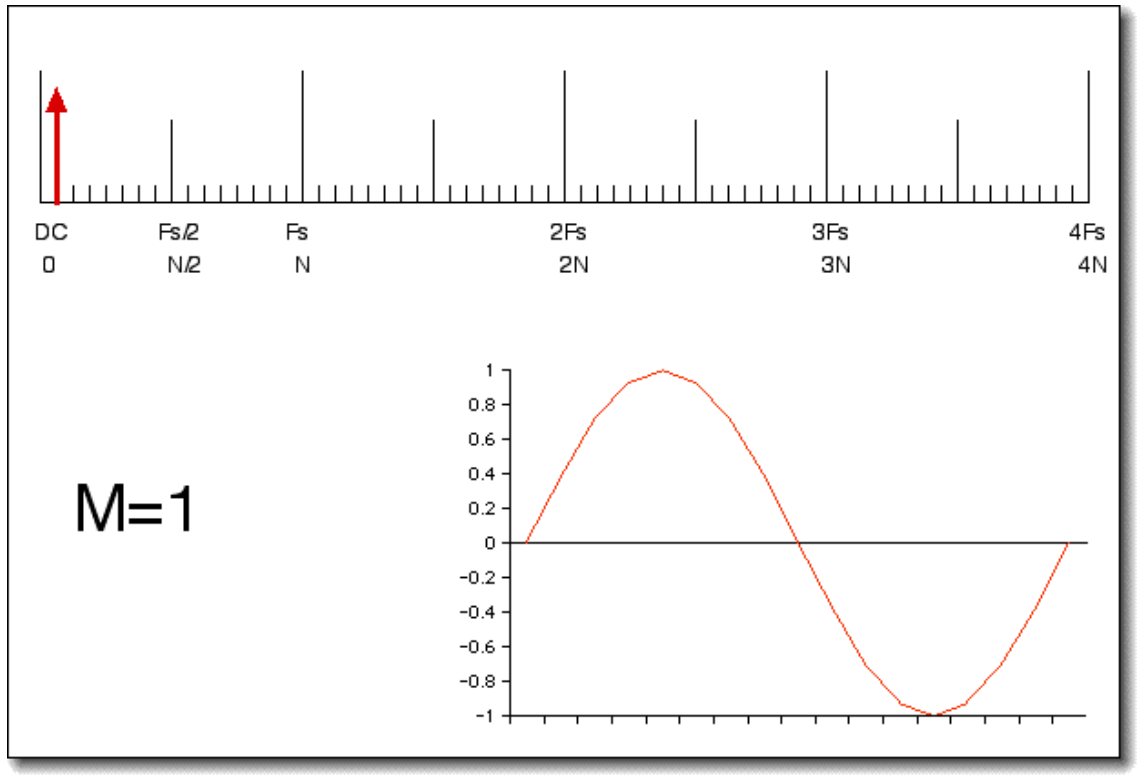


Localization challenges - Sensor Aliasing

- **Aliasing** occurs when a **reconstructed signal** from **samples** of the **original signal** contains **low frequency** components that **are not present** in the **original one.**
- The **cause** is the components at frequency exceeding a certain frequency called Nyquist frequency **Fs/2**, where **Fs** is the sampling frequency (**undersampling**).
- The presence of two components means that the samples can fit at least two different sinusoids, one of which is with the true frequency. Another sinusoid is **with an alias** frequency ***F – Fs***.



Localization challenges - Sensor Aliasing

- The problem **posed** to **navigation** because of **sensor aliasing** is that, even with **noise-free** sensors, the amount of information is generally **insufficient** to identify the robot's **position** from a **single-percept** reading.

- Thus **techniques** must be employed by the robot **programmer** that base the **robot's** localization on a **series** of **readings** and, thus, **sufficient** information to **recover** the robot's **position** over time.